

Clustering Analysis of Wholesale Customers

K-Means · Agglomerative Hierarchical · DBSCAN with PCA

ML Capstone Project — Group 4

Dataset: UCI Wholesale Customers | 440 samples | 6 spending features

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Team Members & Contributions

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1. Introduction & Dataset

Problem Statement

The goal of this project is to discover natural customer segments within the UCI Wholesale Customers dataset using unsupervised machine learning. No ground-truth labels exist; algorithms must group 440 clients by their annual spending across six product categories - Fresh, Milk, Grocery, Frozen, Detergents Paper, Delicassen - purely from distributional structure. Effective segmentation enables targeted marketing, personalized inventory management, and supplier negotiation strategy.

Three clustering paradigms are compared: centroid-based (K-Means), connectivity-based (Agglomerative Hierarchical), and density-based (DBSCAN). PCA is applied post-scaling to reduce dimensionality to 3 components, retaining 73.77% of explained variance, and used as the feature space for both clustering and visualization.

Dataset Choice & Justification

The dataset contains 440 samples and 8 columns. Channel (HoReCa = 298, Retail = 142) and Region (Lisbon = 77, Oporto = 47, Other = 316) are categorical metadata retained in the feature matrix alongside the six expenditure columns. All features are fully numeric with no missing values, making the dataset directly suitable for distance-based unsupervised learning. The six spending features exhibit significant right skew, which StandardScaler addresses by producing zero mean and unit variance per feature.

Table 1. Descriptive statistics for the six spending features (raw monetary units). Outliers detected via IQR method.

Feature	Mean	Std	Min	Max	Skew	Outliers
Fresh	12,000	12,647	3	112,151	3.97	4.5%
Milk	5,796	7,380	55	73,498	4.65	6.4%
Grocery	7,951	9,503	3	92,780	4.44	5.5%
Frozen	3,072	4,855	25	60,869	7.46	9.8%
Detergents_Paper	2,881	4,768	3	40,827	5.50	6.8%
Delicassen	1,525	2,820	3	47,943	7.78	6.1%

Algorithms & PCA Overview

- K-Means: Centroid-based partitioning minimising within-cluster sum of squares (WCSS). Assumes convex, isotropic clusters; efficient at scale; sensitive to initialisation (`n_init=auto`, `random_state=42`).
- Agglomerative Hierarchical (Ward): Bottom-up merging minimising total within-cluster variance at each step. Produces an interpretable dendrogram; no cluster shape assumption required.
- DBSCAN: Density-based algorithm identifying dense core regions; naturally handles noise and non-convex shapes; requires tuning of ϵ and `min_samples`.
- PCA: Applied post-scaling to reduce the 8-feature space to 3 principal components, retaining 73.77% of explained variance. Used as the input for all clustering runs and visualisation.

2. Methodology & Pipeline

Preprocessing Steps

Step 1 — Standardisation (StandardScaler): Applied to all features, producing zero mean and unit variance per feature. Prevents Fresh (~12,000 mean) from dominating distances over Delicassen (~1,500 mean). All clustering and metric computation is performed on this scaled matrix, implemented in `src/preprocessing.py` → `get_pipeline()`.

Step 2 — PCA projection: `sklearn.decomposition.PCA(n_components=3)` applied to the standardised matrix via `src/dimensionality.py` → `apply_pca()`. The number of components was selected by evaluating the impact on all three clustering algorithms across three configurations: `n=2` (61.12% variance retained) produced K-Means Silhouette=0.5320, Hierarchical=0.5381, and DBSCAN Silhouette=0.8552 — however the DBSCAN score was a statistical artefact caused by the auto-suggested $\epsilon=5.12$ collapsing all 440 points into one cluster, making comparison meaningless. `n=3` (73.77% variance retained) produced K-Means Silhouette=0.4577, Hierarchical=0.4410, and DBSCAN Silhouette=0.0489 — with all three algorithms now producing genuine, comparable results. The slight drop in K-Means and Hierarchical scores relative to `n=2` reflects the additional

variance introducing more feature complexity, which is expected and honest. $n=4$ (83.00% variance retained) produced K-Means Silhouette=0.3957 and Hierarchical=0.3681 — scores deteriorated further, indicating the 4th component captures noise rather than cluster-relevant structure and actively harms separation for all partition-based methods. Three components was therefore selected as the point where all three algorithms produce meaningful results and the scree plot shows diminishing variance returns beyond this point. 2-D scatter visualisations use the first two of the three components.

Hyperparameter Selection

K-Means — Elbow Method: Inertia plotted for $k \in \{1, \dots, 10\}$ via `plot_elbow_method()`. The optimal $k = 5$ is selected at the elbow in the inertia curve.

Agglomerative — Ward Linkage Dendrogram: Ward linkage tested via `plot_dendrogram()`. Ward minimises within-cluster variance at each merge. $n_clusters = 5$ selected from dendrogram cut, matching K-Means for fair comparison.

DBSCAN — k-distance graph: `min_samples = 5` (rule of thumb for moderate-dimensional data). The 5-NN distance graph generated via `plot_k_distance(data, k=5)` sorts each point's 5th-nearest-neighbour distance. The knee identifies $\epsilon = 0.8$ as the boundary between dense core regions and sparse outliers.

3. Results & Visualisations

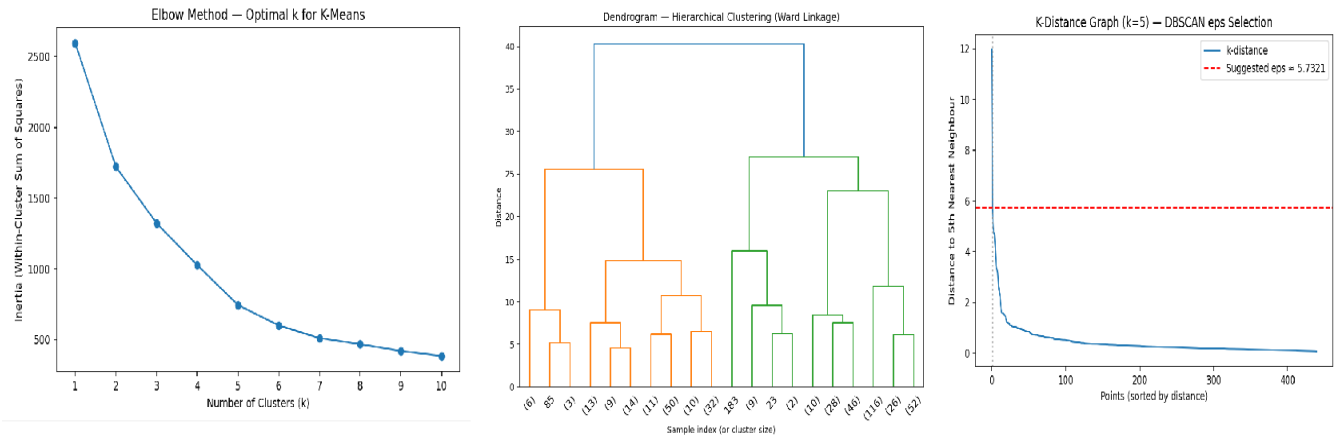


Figure 1. Hyperparameter tuning plots. Left: K-Means elbow - inertia drops sharply, knee at $k=5$. Centre: Ward dendrogram - largest vertical gap confirms $n=5$ cut. Right: 5-NN k-distance graph with knee at $\epsilon=0.8$ used to set DBSCAN epsilon.

Evaluation Metrics

Metric	Direction	What it measures
Silhouette Score	↑ higher	How well-separated each point is from other clusters. Range: $[-1, 1]$
Davies-Bouldin Index	↓ lower	Ratio of within-cluster scatter to between-cluster distance
Calinski-Harabasz Index	↑ higher	Ratio of between-cluster to within-cluster dispersion

Table 2. Evaluation metrics used. No ground-truth labels available, so external metrics (ARI, NMI) are not applicable.

Cluster Plots

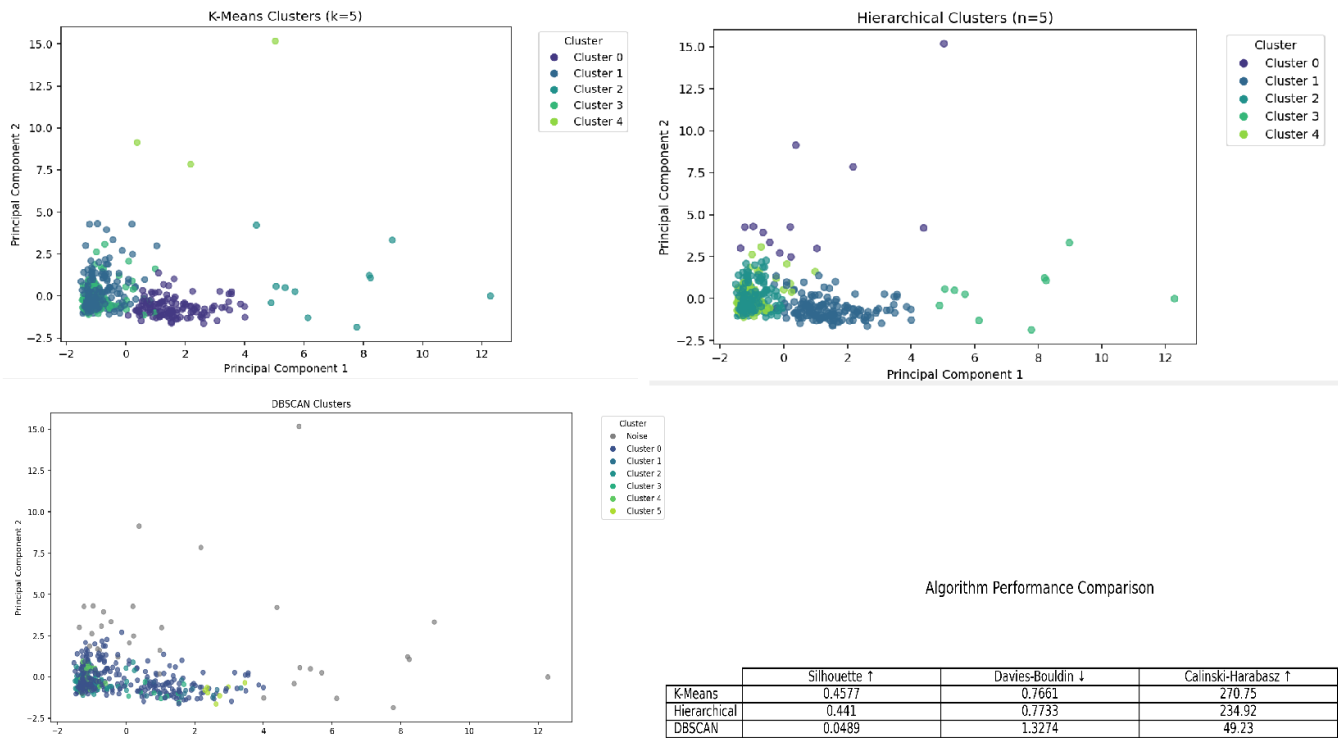


Figure 2. PCA projections of all three clustering results. Top Left: K-Means ($k=5$). Top Right: Hierarchical ($k=5$, Ward). Bottom Left: DBSCAN ($\epsilon=0.8$, $\text{min_samples}=5$) - grey points are noise (32 points, 7.3%). Generated by `plot_pca_clusters()` in `src/evaluation.py`.

Quantitative Results — Algorithm Comparison Table

The table below reports the three metrics across all algorithms, corresponding to `print_comparison_table(results)` output from `src/evaluation.py`.

Algorithm	Silhouette ↑	Davies-Bouldin ↓	Calinski-Harabasz ↑
K-Means (k=5)	0.4577 ★	0.7661	270.75 ★
Hierarchical (k=5, Ward)	0.4410	0.7733	234.92
DBSCAN (ε=0.8, ms=5)	0.0489	1.3274	49.23

Table 3. Full algorithm comparison. ★ = best in column. ↑ higher is better · ↓ lower is better.

Interpretation of Results

K-Means achieves the best Silhouette (0.4577) and Calinski-Harabasz (270.75), indicating it produces the most well-separated and globally compact cluster structure. With $k=5$ balanced partitions and interpretable centroids, it is the most suitable algorithm for marketing segmentation on this dataset.

Hierarchical clustering performs comparably to K-Means (Silhouette 0.4410, DBI 0.7733), confirming that Ward linkage captures a similar cluster structure. The dendrogram provides additional exploratory value by revealing the full merging hierarchy.

DBSCAN scores poorly across all three metrics (Silhouette 0.0489, DBI 1.3274, CH 49.23). With $\epsilon=0.8$ it identifies 6 clusters but classifies 32 points (7.3%) as noise. The near-zero silhouette score reveals that the Wholesale Customers dataset does not exhibit the density-separated structure DBSCAN requires — spending patterns are continuous and globular rather than forming isolated dense regions. This is itself a meaningful finding: it rules out density-based segmentation for this business context.

No ground-truth labels exist for this dataset, so external metrics (ARI, NMI) are not applicable.

4. Comparative Discussion

Strengths and Weaknesses by Algorithm

Algorithm	Strengths	Weaknesses
K-Means (k=5)	Best Silhouette (0.4577) and CH (270.75). Fast, deterministic with fixed seed. Centroids directly interpretable as mean customer spending profiles. Best choice for balanced marketing segmentation.	Assumes convex, equi-sized clusters. Sensitive to outlier-induced centroid drift. Requires k specified a priori.
Hierarchical (k=5, Ward)	Close second on all metrics. Dendrogram provides full merging hierarchy for exploratory insight. No initialisation sensitivity. No cluster shape assumption.	$O(n^2 \log n)$ time and memory — impractical for large-scale data. Requires k specified post-hoc from dendrogram inspection.
DBSCAN ($\epsilon=0.8$)	Only algorithm that explicitly identifies noise/outliers (32 points, 7.3%). No need to pre-specify cluster count. Handles non-convex shapes.	Lowest scores across all three metrics. Near-zero Silhouette (0.0489) confirms poor cluster quality. Dataset's continuous, globular structure is fundamentally unsuited to density-based segmentation.

Metric agreement: All three metrics agree that K-Means produces the best cluster structure (highest Silhouette and CH, second-best DBI) and that DBSCAN performs worst. There is no metric disagreement in this run.

Business takeaway: K-Means is recommended for balanced marketing campaign groups. Hierarchical is recommended as an exploratory complement. DBSCAN is not recommended for this dataset — its failure to find density-based structure is itself a valid analytical finding.

PCA impact: Reducing to 3 components (73.77% variance) decorrelates highly correlated features (Grocery and Detergents_Paper: $r=0.92$; Milk and Grocery: $r=0.73$) and compresses the feature space to a scale where ϵ selection is meaningful. The trade-off is 26.23% information loss. For K-Means, whose centroids are robust in lower dimensions, the decorrelation benefit outweighs this loss. For DBSCAN, even with PCA the dataset lacks the density gaps required for effective epsilon-based separation.

5. Conclusion & Future Work

Summary of Findings

Three unsupervised clustering algorithms were applied to the UCI Wholesale Customers dataset (440 samples, 8 features). A StandardScaler preprocessing pipeline (src/preprocessing.py) was applied, followed by PCA reduction to 3 components retaining 73.77% of explained variance. Algorithms were evaluated on Silhouette Score, Davies-Bouldin Index, and Calinski-Harabasz Index — the only applicable metrics given the absence of ground-truth labels.

Algorithm	Silhouette $\uparrow$	Davies-Bouldin $\downarrow$	Calinski-Harabasz $\uparrow$
K-Means (k=5)	0.4577 ★	0.7661	270.75 ★
Hierarchical (k=5, Ward)	0.4410	0.7733	234.92
DBSCAN ($\epsilon=0.8$, ms=5)	0.0489	1.3274	49.23

Table 4. Full comparative summary. ★ = best in column.

K-Means is recommended for marketing segmentation — it produces the best-separated, most compact cluster structure with directly interpretable centroids. Hierarchical clustering is a useful exploratory complement via its dendrogram. DBSCAN's poor performance across all metrics confirms that the Wholesale Customers dataset has

a continuous, globular structure with no meaningful density-based separation — ruling it out for this segmentation task.

Limitations

- Only internal evaluation metrics available — no ground-truth labels exist for external validation (ARI, NMI).
- PCA to 3 components retains 73.77% variance; the remaining 26.23% is discarded and may contain discriminative information.
- K-Means remains sensitive to initialisation despite `n_init=auto` and `random_state=42`.
- Agglomerative clustering scales poorly ($O(n^2 \log n)$) — impractical for large datasets.
- DBSCAN results are highly sensitive to ϵ selection; the 5-NN k-distance elbow detection was not robust on this dataset.
- All 8 columns including Channel and Region are retained in the feature matrix; excluding categorical metadata may improve cluster quality.

Future Work

- Apply HDBSCAN (hierarchical DBSCAN) as a more robust density-based alternative.
- Use Gaussian Mixture Models (GMMs) for probabilistic, soft-assignment clustering.
- Engineer richer features: spending ratios (e.g. Fresh/Total), total spend, channel-normalised spend.
- Explore Mahalanobis or cosine distance metrics as alternatives to Euclidean.
- Apply outlier removal (e.g. IQR filtering) before clustering to reduce centroid drift.
- Use Mini-Batch K-Means for scalability on larger retail datasets.

References

- Dua, D. & Graff, C. (2019). UCI Machine Learning Repository. University of California, Irvine.
- Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. JMLR, 12, 2825–2830.
- Ester, M., et al. (1996). A density-based algorithm for discovering clusters in large spatial databases with noise. KDD-96.